

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

Schism: Splitting RV32 from `arch/riscv`

We Could Both Be Better as We Part Ways



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Who am I ?

Ruinland,

Employed by Andes Technology.

Currently humbly serving as the chair of System Library Working Group for RISE,







Agenda

- Splitting RISC-V 32 Linux in technical aspect, how ?
- Just, why and what is proposed, exactly ?
- And what we want to do after that ?
- Who will maintain it and what the responsibility is ?

How ?

Well, pretty much just

1. Copy and clean `arch/riscv` into `arch/riscv32`.
2. Fix a bunch of hard-coded path in Makefiles and Kconfigs/Kbuilds.
3. Cleanup the remaining of 64bit definitions. (I got lazy here.)
4. `make arch=riscv32 Image`
5. Profit !

```
# OBJCOPY arch/riscv32/boot/Image
  /home/ruinland/machdyne_lakritz/buildroot/output/host/bin/riscv32-linux-objcopy  -O binary -R .note -R .note.gnu.build-id -R .comment -S vmlinux arch/riscv32
/boot/Image
```

The end (mic drop)

Oh dear, of course I am kidding.

😛😛😛

Just, why ?

Chicken-egg dilemma of spec and implementation - -

Open source developers asked for a stable base to target at.

For instance, Rich Felker questions that whether RISC-V 32 Linux has a stable ABI :

https://www.openwall.com/lists/musl/2020/03/12/2

=> RISC-V 32 doesn't have a proper profile. RVA and RVB are only for 64-bit platforms.

=> Developers doesn't want to waste time on things might change, just like the RVV 0.7 fiasco we had.

=> No devs, no one wants to talk about it.

=> No one pulling together the a spec.

Also, many extensions are ignoring RV32 while the RV32 and 64 code reside together.

A vicious cycle that drains the possibilities.

What we are proposing, exactly ?

Why vendors didn't propose a RVA profile ?

Vendors who uses RV32 Linux are extremely highly diverse and sometimes have very peculiar designs.

Now we have RVB - - B stands for "bespoke."

What we are proposing : To establish a **minimal and sane RV32 profile** and get the code base maintained separately.

And what we want to accomplish with that RVB profile?

- To clarify that RV-A isn't always necessary : kernel assisted userspace atomicity.

SuperH GUSA (General UserSpace Atomicity) and ARM pre-V5 Kernel Provided User Helpers .

Patches are among glibc, gcc (libatomic), and kernel. (will be posted after fully polished.)

- By clarifying max usable memory, re-introduce RV32 HIGHMEM support.
- To clarif that MMU isn't always necessary, either and keep noMMU support maintained.
- ..... any features requiring a spec to back it up.

Kernel provided user helpers

Still Work in Progress - -

- **glibc : sysdeps/riscv/atomic-machine.h**

  Internal locks and atomic logics

- **gcc :**
  - **libgcc/config/riscv/atomic.c**
  - **libatomic/config/linux/riscv-config.h**

  So as to provide __builtint_atomic_*()

- **kernel : lay the logics in arch/riscv32/entry.s**

  Provide the ARM pre-v5 styled region of code for direct invoking. No vDSO because for considering noMMU platforms.

Look and feel from the real world :

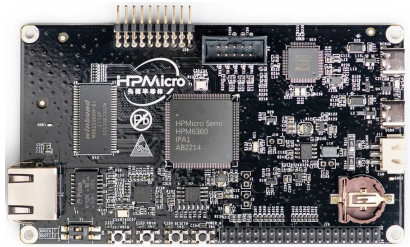
```
245 android_atomic_cmpxchg:
246     stmdb   sp!, {r4, lr}
247     mov     r4, r0          /* r4 = save oldvalue */
248 1: @ android_atomic_cmpxchg
249     mov     r3, #kernel_atomic_base
250 #ifdef __ARM_HAVE_PC_INTERWORK
251     add     lr, pc, #4
252     mov     r0, r4          /* r0 = oldvalue */
253     add     pc, r3, #(kernel_cmpxchg - kernel_atomic_base)
254 #else
255     mov     r0, r4          /* r0 = oldvalue */
256     add     r3, r3, #(kernel_cmpxchg - kernel_atomic_base)
257     mov     lr, pc
258     bx      r3
259 #endif
260     bcs     2f              /* swap was made. we're good, return. */
261     ldr     r3, [r2]        /* swap not made, see if it's because *ptr!=oldvalue */
262     cmp     r3, r4
263     beq     1b
264 2: @ android_atomic_cmpxchg
265     ldmia   sp!, {r4, lr}
266     bx      lr
```

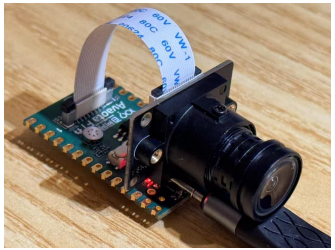
ref :
https://android.googlesource.com/platform/system/core/+/dd7bc3319deb2b77c5d07a51b7d6cd7e11b5beb0/libcutils/atomic-android-arm.S#245

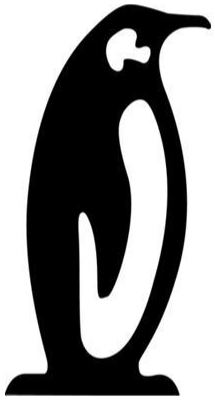
Who will maintain it and what the responsibility is ?

Architecturally (?) speaking, Andes is the one and only RISC-V IP vendor who sells RV32 Linux capable IPs and get it mass produced : to name a few, Allwinner V821 and HPMicro HPM6 series.

We are more than happy to lift the burden and are welcoming others to establish RVB-32 (what a coincidence with RVA32).

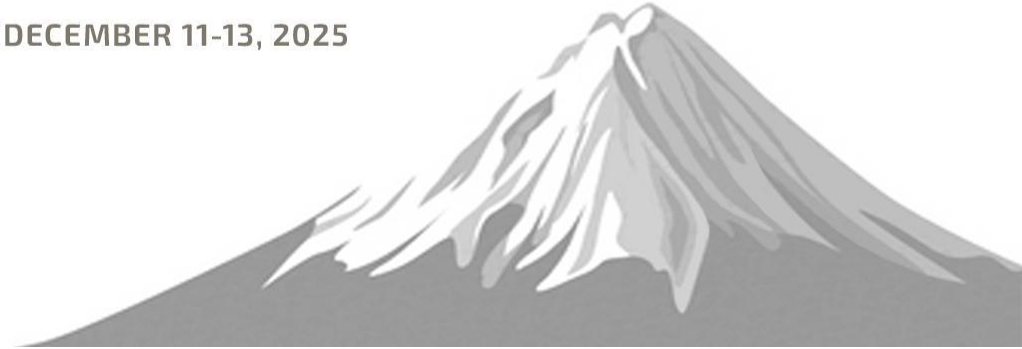






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Backup

Place holders

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